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Optimal Wells Placement to Maximize the Field Coverage Using Derivative-Free Optimization

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Abstract

Mining optimization in oil and gas industry has been an important task. The production volume of oil and gas crucially depends on many factors. One of them is wells location. This study contributes to the domain of oil and gas exploration and extraction. We solve the problem of optimal wells placement on 2-D surface. We have developed a set of criteria and chosen a proportion of the field covered as objective function. In order to find the best solution satisfying these criteria, we compared different optimization algorithms. Such as, Nelder-Mead method, simulated annealing, Bayesian optimization and genetic algorithm in order to reach the maximal surface coverage with minimal number of wells in a region. To test the application of the developed algorithm on complex polygons (oil fields), we have made a generator of synthetic polygons. In this study we also compare the efficiency of regular and irregular wells placement.

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Keywords: optimal wells placement, wells location, stochastic optimization, bayesian optimization, genetic algorithm, nelder-mead, simulated annealing

1. Introduction

In order to reach the sustain production of oil and gas, huge industrial companies implement intellectual systems that help to optimize the production on every stage. For a large oil field, the problem of wells placement is non-linear,

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nonconvex and expensive in terms of computations. The problem of optimal well placement is not new and there are many studies in the literature, developing and comparing different approaches. They can be divided into two groups: gradient based and derivative-free methods. More details on previous studies will be provided in chapter 2. In our study we focus on derivative-free approaches, such as simulated annealing, Bayesian optimization, evolutionary computation and Nelder-Mead algorithm. The methods are described in Methodology section. We also formulate some limiting criteria for wells placement and use it as parameters in methods. As objective function the proportion of covered field is used. The optimization algorithms are applied to determine the coordinates of well's centers and the angle of wells. In addition, we compare two types of placement: regular and irregular (fig.1). They will also be discussed in detail in methodology section.

2. Related works

Previous research works in the field of oil production are dedicated to optimization of wells location [1,2,26,27] and well controls [3-5]. Some of the studies investigate the optimal number of wells [6]. Sometimes, researchers combine several tasks, for instance: locations and controls [7].

The existing studies also differ in terms of the objective function: some eager to place wells in order to maximize the production level [8], others are aiming to reach the maximal net present value [9]. There are also studies dedicated to the minimization of particular effects while locating the wells. For instance, minimization of risks of sulphate scale deposition [10]. Concerning the objective function, in our study, we maximize the coverage proportion. The present study is divided into 2 stages. On the first stage of research, described in the present paper, we are aiming to maximize the surface coverage due to the limitations to the access of the production volume data. In the second part, we will combine the results with hydrodynamic model in order to maximize the production level. To our knowledge, the task of surface coverage optimization in wells placement is poorly covered in academic literature and may be enriched with our work.

The problem of wells location can be solved using gradient-based algorithms [11-12]. For instance, simultaneous perturbation stochastic approximation algorithm [13]. Another study is applying the steepest ascent method However, since the solution space is nonconvex and not always continuous, the derivatives can hardly be calculated. Therefore, many studies apply derivative free methods to the task [9].

Another direction of research lies in the application of derivative-free methods to the task of optimal wells location. Particularly, evolutionary algorithms are widely used in the industry, especially for wells placement [14-17]. Another common optimization algorithm is simulated annealing. It has been applied to the placement task since 1995 by Beckner and Song and have become one of the first solutions to the wells' optimal location [18]. Researchers also tend to use Nelder-Mead algorithm to the task and results perform a positive effect [19]. A more recent algorithm used for the task is Bayesian optimization [20-21].

3. Methodology

3.1 Task setting

We define two types of wells placement: regular and irregular. By regular placement we understand the setting, when all wells have fixed angle of drainage area and fixed distance between wells. On opposite, in irregular placement, wells may have different drainage area angle and different inter-well distance. The purpose of development an irregular placement is to let the well's more smoothly adopt to the shape of the oil field (Fig. 1).

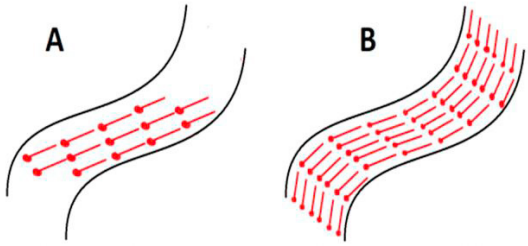

Figure 1 Comparison of regular and irregular wells placement

In the regular placement with fixed angle for all wells parameters we are generating a net of wells, with fixed interwell distance and fixed angle for all wells in the net. The optimized parameters are the coordinates of the first well in the net and the angle of all wells in the net.

In the irregular setting we propose to divide the filed in parts and find the best placement parameters within these parts. As a separator we use a line, which location is being optimized together with the placement parameters (coordinates of the first well and the angle for the well in each part of the field). Our goal is to check, which type of the setting can help to reach the maximal surface coverage&

3.2 Formalization of placement limiting criteria

3.2.1 Minimal distance between wells' centres

Let C be a set of center coordinates for all wells in a set W_{-i}, except well w_i. The distance d_{wells_c} between well's w_i center $\left(x_{w_j}, y_{w_j}\right)$, and other wells from the C (x_{C_i}, y_{C_i}) must be no more than the minimum allowable distance $d_{min_{wells_c}}$. Distance d_{wells_c} can be calculated as the length of a segment with borders at points (x_{w_j}, y_{w_j}) и (x_{C_i}, y_{C_i}) with formula 1:

$$d_{wells_c} = \sqrt{(x_{w_j} - x_{c_i})^2 + (y_{w_j} - y_{c_i})^2} \quad (1)$$

From this we obtain that the criterion of the minimum permissible distance between the coordinates of the centers of the wells is determined as follows:

$$d_{wells_c} < d_{min_{wells_c}},$$

где d_{wells} - the actual distance from the side of the drainage zone $ABCD_{w_j}$ of the well w_j to each side of the drainage zones of other wells from the set Q ; $d_{min_{wells_C}}$ – the minimum allowable distance between the centers of the wells from the set (Fig. 2.3).

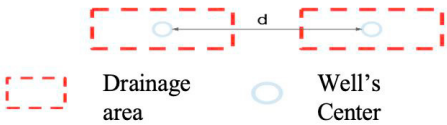

Figure 2 Visualization of minimal permissible distance between wells.

3.2.2 Criterion for the minimum proportion of wells crossing the border.

Suppose, some wells can cross the boundaries of a field, as shown in the figure 3.

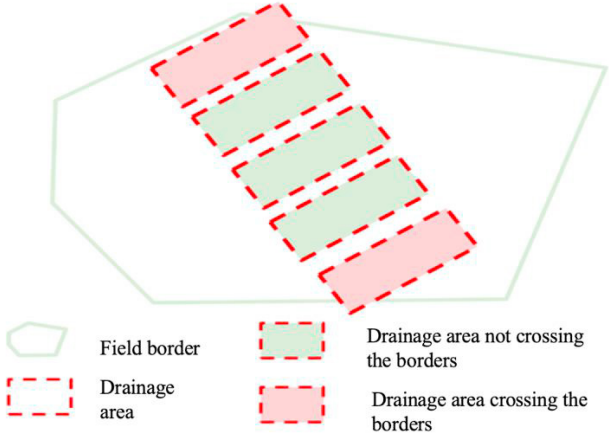

Figure 3 An example of the arrangement of wells with the crossing of drainage zones beyond the boundaries of the field

The number of wells crossing the border, N_{out} with respect to the total number of wells N must be less than the permissible maximum value $N_{out_{max}}$:

$$\frac{N_{out}}{N} < N_{out_{max}}$$

3.3 Formalization of wells placement objective criteria

3.3.1. *Proportion of field coverage*

The main quality criterion is the fraction of the sum of all surface area of drainage zones of all wells located within the boundaries of the field S_{ABCDin}, in relation to the area of the oil field S_M.

$$l_S = \frac{\sum_{i=1}^{N} s_{ABCD_{in_i}}}{S_M} \rightarrow \boldsymbol{max}$$

where N- number of wells, S_M – oil field surface area, $s_{ABCD_{in_i}}$ – a surface are of the i-th well, located inside the oil field. The value of l_S should be maximized. The maximal possible value of l_S is 1.

3.3.2. *The ratio of the number of wells used to cover the field to the "optimal number of wells":*

$$l_N = \frac{\mathrm{N}}{N_0} \rightarrow \boldsymbol{min}$$

where N- number of wells, N_0 – optimal number of wells.

In this case, the "optimal number of wells" is defined as a real number equal to the ratio of the area of the field to the area of the drainage zone of one field:

$$N_0 = \frac{S_M}{s_{ABCD}}.$$

Where S_M – oil field surface area, $s_{ABCD_{in_i}}$ – a surface are of the i-th well, located inside the oil

This criterion is to some extent the opposite of the previous one. With arrangements close to optimal l_N will be close to 1.

3.3.3. Combined criteria

In the best scenario, the maximum coverage of the field area should be realized with a minimum number of wells. This requires combining the two previous criteria. We can do this as follows

$$l_{Comb} = l_S + \frac{\epsilon}{l_N} \rightarrow \boldsymbol{max},$$

where ε – coefficient selected on the basis of economic calculations,

3.4 Optimization algorithms

Our approach to the solution in regular setting, undermines the generation of wells' net with fixed angle of drainage area and fixed distance between wells. In terms of optimization, the task of wells placement comes down to calculation of optimal parameters of the net: (x_{start}, y_{start}) - start well coordinates, α – angle of rotation of wells.

3.4.1 *Nelder-Mead Algorithm*. A method of optimization (searching for a local minimum) for a function of several variables. The algorithm allows optimizing functions without using gradients. It consists in the formation of a simplex and its subsequent deformation in the direction of the minimum, through three operations: reflection, expansion and contrast. Each operation is performed sequentially if a new best point was not found in the previous step [22].

3.4.2 *Evolutionary computation*. Evolutionary modeling is a field in artificial intelligence based on principles and a conceptual framework borrowed from evolutionary biology and population genetics and combining computer methods (genetic algorithms, genetic programming, evolutionary programming and evolutionary strategies) for modeling evolutionary processes in artificial systems [23].

We used an Evolutionary computation framework from inspyred Python 3.6 library with the following parameters (inspired.ec module): max_evaluations=200, num_selected=20, mutation_rate=0.25, number of iterations = 10.

3.4.3 *Bayesian Optimization*. The goal of Bayesian optimization is to build a distribution $P(y/x)$, based on previous measurements, a priori information) using a Gaussian random process. We use and An algorithm from open source bayes_opt Python 3.6 library [24] with the following parameters: number of initial points = 1, number of iterations = 5.

3.4.4 *Simulated annealing*. Stochastic global maximum search algorithm, suitable for nonsmooth functions is also compared with other methods [25]. Maximal number of iterations set to 50, due to the fast computing, the parameter of temperature is set to 0.1.

4. Experiments

We conducted a series of experiments using synthetic oil fields in order to evaluate the effectiveness of irregular arrangement (when dividing with a straight line) compared to a regular one. Random boundary and Y-shaped polygons were created by the random field boundary generator. By Y-shaped polygons we mean polygons with the presence of at least two narrow enough parts to fit in at least one well. It was made to create more complex conditions for optimizers (examples in figure 4).

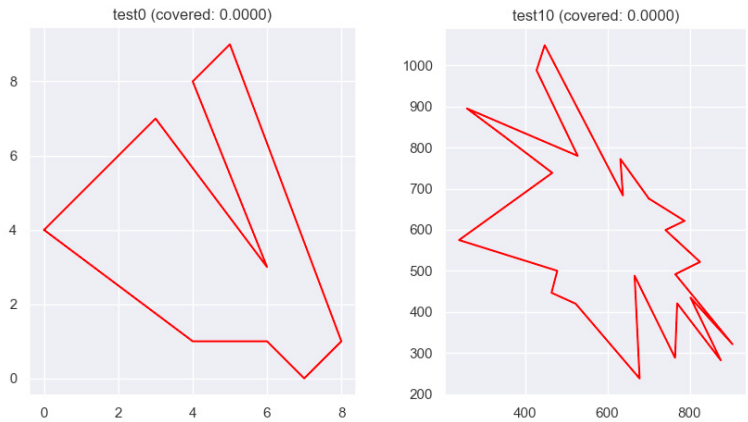

Figure 4 On the right: Synthetic field with Y-shape. On the left: Synthetic field with random shape

For the experiments, 30 random and 10 Y-shaped polygons were generated. The largest number of iterations was provided to simulating annealing due to the high speed of calculations, the Bayesian optimizer, on average, converged to a locally optimal solution in 5 iterations, the evolutionary algorithm was limited to ten iterations.

In addition to generated fields, the algorithms were also tested on one real example of the oil field.

5. Results

Concerning the regular setting, the placement results on real field are provided in figure 5. The highest coverage score was achieved by Nelder-Mead algorithm. Evolutionary computation and simulated annealing scored similar performance and are also close to Nelder-Mead in terms of coverage.

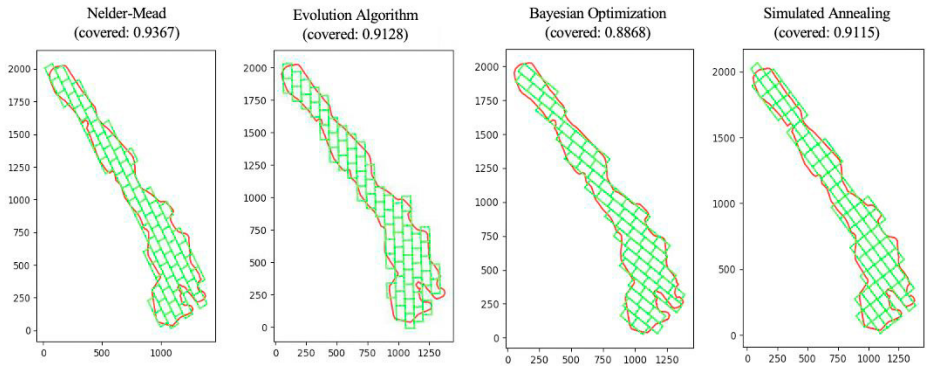

Figure 5 Examples of wells placement by different optimization algorithms on the real field

However, Nelder-Mead algorithm strongly depends on the parameters initialization values which can be hard for a user to set. Therefore, it doesn't work stably and we decided to exclude it from the further experiments.

Since we have a lack of real field examples, we decided to enrich the dataset with synthetic data for further testing. The results are presented in graphs 6-8. The results for regular and irregular arrangements for identical polygons are shown here. A quality metric is displayed on a vertical axis, which we maximize.

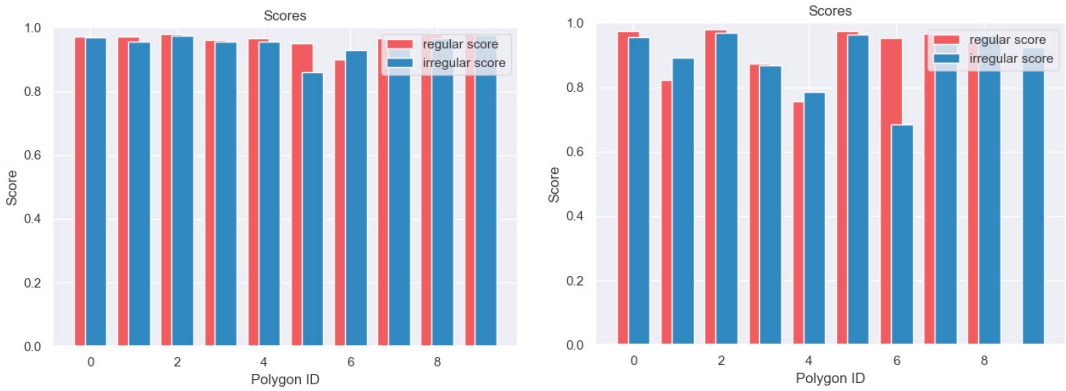

Figure 6 Evolutionary computation coverage score for regular and irregular optimization (Left – random shape fields, right – Y-shape fields)

The evolutionary algorithm of Fig. 2.32 in most cases works better with regular placement. Moreover, in most cases, an irregular arrangement is slightly inferior to a regular one. However, there are cases of polygons for which only irregular optimization is suitable example: polygon 10 fig. 7 right. This effect, according to observations, occurs when the landfill has an elongated and narrow section in which it is impossible to fit the well and the "unsuccessful'

initial parameters of some optimizers.

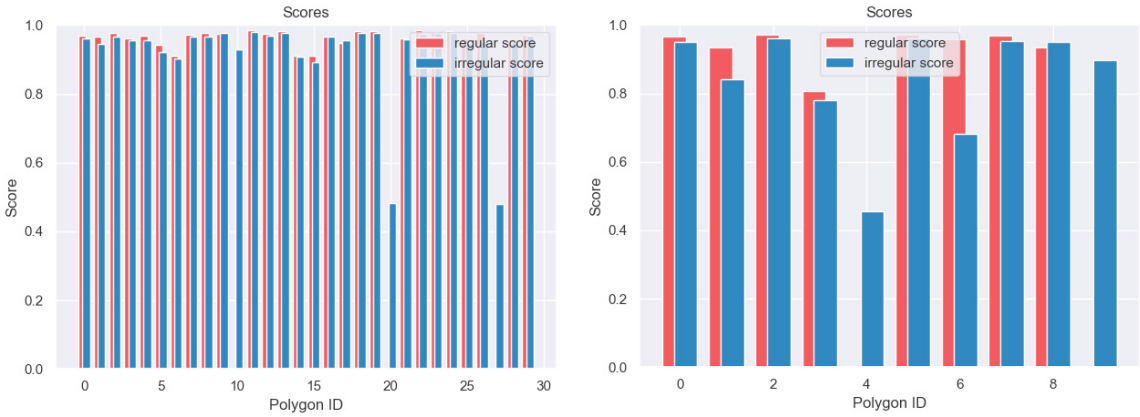

Figure 7 Bayesian optimisation coverage score for regular and irregular optimization (Left – random shape fields, right – Y-shape fields)

The Bayesian optimizer also works better with regular placement, but also for some polygons where the regular placement did not work, an irregular placement may be better due to the division of the polygon into parts.

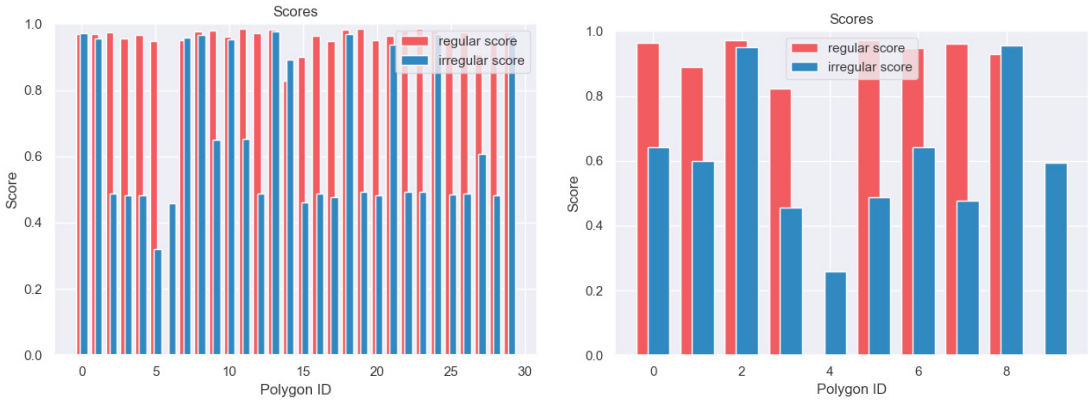

Figure 8 Simulated annealing optimisation coverage score for regular and irregular optimization (Left – random shape fields, right – Y-shape fields)

Simulated annealing (Fig. 8) works much more efficiently with regular placement. In most cases, it is twice as large as irregular. We do not recommend using it for irregular arrangements when dividing the polygon into parts, but it performs good and stable results for the regular placement.

Table 1. Computational time of optimization algorithm for regular placement.

Method	Minimal Time, seconds	Maximal Time, seconds
Evolutionary Computation	10	350
Simulated Annealing	0.5	9
Bayesian Optimization	4	30

We also compared the running time of the algorithms. All methods, on average, work much faster with regular placement. The fastest algorithm for problems of regular and irregular arrangement turned out to be simulated annealing, then a Bayesian optimizer and an evolutionary algorithm. The evolutionary algorithm is the slowest of all: on regular constellations, the operating time takes from 10 to 350 seconds. On irregular: from 40 to 1200 seconds. The Bayesian algorithm on regular arrangements worked from 5 to 30 seconds. On irregular: from 120 to 750 seconds. Simulated annealing is the fastest: regular placement takes from 0.5 seconds to 9 seconds, irregular - from 1 to 13 (fig.9).

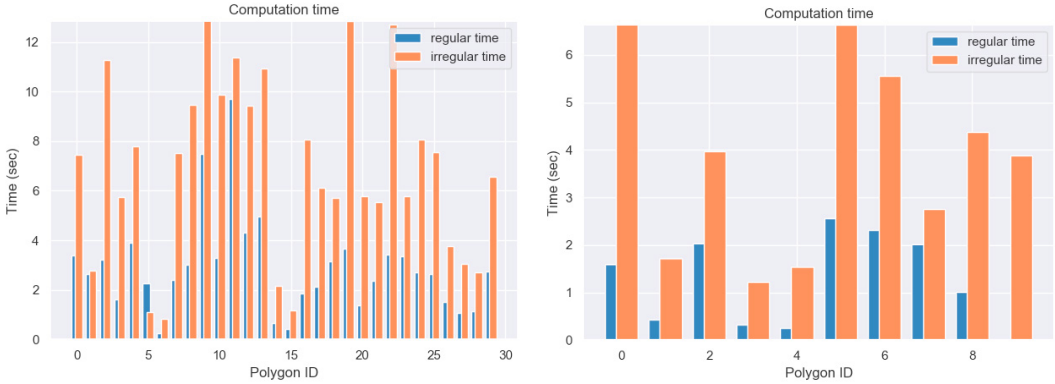

Figure 9 Simulated annealing. Computational time for regular and irregular wells placement

According to the results of the experiments, it can be said that in most cases regular well placement is more efficient in terms of coverage proportion than irregular. The fastest algorithm is simulated annealing.

6. Discussion

Sometimes an irregular arrangement exceeds the regular results (in the case of the evolutionary algorithm), but the difference in these points is not significant. Also, in contrast to a regular arrangement, which stably demonstrates a coverage level of more than 80% for the evolutionary algorithm and more than 65% for simulating annealing, irregular arrangement can work unstably, choosing significantly worse solutions (30% coverage). In addition, the irregular arrangement is much worse to the regular in time work.

Conclusion

In conclusion, we would like to say, that during the research work we have developed and formalized the constraints for wells placement as well as the objective function. We proposed and approach to irregular wells placement using a polygon separation with line. We applied and compared four optimization algorithms for regular and irregular wells placement in terms of coverage and computational time. To test our ideas, we developed a class of oil field generation. Concerning the comparison of regular and irregular wells placement, the regular wells location is more efficient in terms of coverage. Moreover, it works mush faster. Concerning the choice of the optimization algorithm in regular case Bayesian optimization, simulated annealing and evolutionary computations performs close to maximum results, but in terms time, simulated annealing is the best one.

As a result, the wells placement optimization Python 3.6 module is developed. All four algorithms and data generator are implemented and can be used by practitioners.

Acknowledgements

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References

[1] Bouzarkouna, Z., Ding, D.Y. & Auger, A. Well placement optimization with the covariance matrix adaptation evolution strategy and meta-models. Comput Geosci 16, 75–92 (2012).

[2] Alzahabi, Ahmed & Alqahtani, Ghazi & Spinner, Timothy & Soliman, Mohamed. (2014). A Computational Comparison between Optimization Techniques for Wells Placement Problem: Mathematical Formulations, Genetic Algorithms and Very Fast Simulated Annealing. Journal of Materials Science and Chemical Engineering. 2. 59. 10.4236/msce.2014.210009.

[3] Asadollahi, M., Nævdal, G., Dadashpour, M., Kleppe, J., "Production optimization using derivative free methods applied to Brugge field case". Journal of Petroleum Science and Engineering, 114, 2014, pp.22-37,

[4] Oliveira, D. F., Reynolds, A., "An adaptive hierarchical multiscale algorithm for estimation of optimal well controls". SPE Journal, 19 (05), 2014, pp.909-930,

[5] Suwartadi, E., Krogstad, S., Foss, B., "Adjoint-based surrogate optimization of oil reservoir water flooding". Optimization and Engineering, 16 (2), 2014, pp.441-481

[6] Abukhamsin A.,,"Optimization of Well Design and Location in a Real Field" a Report submitted to the department of energy resources engineering. Stanfor university, (2009)

[7] Humphries, T. D., Haynes, R. D., James, L. A., "Simultaneous and sequential approaches to joint optimization of well placement and control". Computational Geosciences, 18 (3-4), 2013, pp.433-448

[8] Wang, C., Li, G., & Reynolds, A. C. (2007)., "Optimal Well Placement for Production Optimization. Society of Petroleum Engineers." doi:10.2118/111154-MS

[9] Wang, Xiang & Feng, Qihong & Haynes, Ronald. (2015). Optimization of Well Placement and Production for Large-scale Mature Oil Fields. Journal of Engineering Science and Technology Review. 8. 134-140. 10.25103/jestr.085.19.

[10] Hutahaean, J. J., Demyanov, V., Arnold, D., & Vazquez, O. (2014, November 10). Optimization of Well Placement to Minimize the Risk of Scale Deposition in Field Development. Society of Petroleum Engineers. doi:10.2118/171733-MS

[11] Zhang, L., Zhang, K., Chen, Y., Li, M., Yao, J., Li, L., Lee, J., "Smart well pattern optimization using gradient algorithm". Journal of Energy Resources Technology, 138 (1), 2015, pp.012901.,

[12] Jansen, J. D., "Adjoint-based optimization of multi-phase flow through porous media–a review". Computers & Fluids, 46 (1), 2011, pp.40-51

[13] Li, L., Jafarpour, B. & Mohammad-Khaninezhad, M.R. A simultaneous perturbation stochastic approximation algorithm for coupled well placement and control optimization under geologic uncertainty. Comput Geosci 17, 167–188 (2013). https://doi.org/10.1007/s10596-012-9323-1

[14] Onwunalu JE (2006) Optimization of Nonconventional well placement using genetic algorithms and statistical proxy. Master's Thesis Report, Energy Resources Engineering Department, Stanford University, USA.

[15] A Comparison of Common Methods for Optimal Well Placement Jeremy J. Minton Supervised by: Associate Professor Rosalind Archer Department of Engineering Science, Faculty of Engineering, University of Auckland, New Zealand.

[16] Montes, G., Bartolome, P., and Udias, A. (2001). The use of genetic algorithms in well placement optimization. In SPE Latin American and Caribbean Petroleum Engineering Conference, Tupac, Y., Almeida, L., Almeida, L., and Vellasco, M. (2007). Evolutionary optimization of oil field development. In Digital Energy Conference and Exhibition, Houston, Texas, U.S.A.,

[17] Morales, A., Nasrabadi, H., and Zhu, D. (2011). A new modified genetic algorithm for well placement optimization under geological uncertainties. In SPE EUROPEC/EAGE Annual Conference and Exhibition, Ding, D. (2008). Optimization of well placement using evolutionary algorithms. In SPE Europec/EAGE Annual conference and exhibition

[18] Beckner, B.L., Song, X. (1995). Field development planning using simulated annealing - optimal economic well scheduling and placement, SPE Annual Technical Conference and Exhibition Dallas, Texas.

[19] Badru, O. (2003). "Well-Placement Optimization Using the Quality Map Approach". Master's Thesis Report, Energy Resources Engineering Department, Stanford University, USA.

[20] Wang, S., & Chen, S. (2017, October 9). A Novel Bayesian Optimization Framework for Computationally Expensive Optimization Problem in Tight Oil Reservoirs. Society of Petroleum Engineers. doi:10.2118/187284-MS

[21] Dou, Zengyi (2015), "Bayesian global optimization approach to the oil well placement problem with quantified uncertainties" . Open Access Theses. 530.https://docs.lib.purdue.edu/open_access_theses/530

[22] J. A. Nelder, R. Mead, A Simplex Method for Function Minimization, The Computer Journal, Volume 7, Issue 4, January 1965, Pages 308–313, https://doi.org/10.1093/comjnl/7.4.308

[23] A.E. Eiben, M. Schoenauer (2002), "Evolutionary computing". Information Processing Letters 82, 1–6

[24] Nogueira F., (2014) "Bayesian Optimization: Open source constrained global optimization tool for Python", https://github.com/fmfn/BayesianOptimization

[25] Kirkpatrick, Scott & Gelatt, C. & Vecchi, M.. (1983). Optimization by Simulated Annealing. Science (New York, N.Y.). 220. 671-80. 10.1126/science.220.4598.671.

[26] Simonov, M., Shubin, A., Penigin, A., Perets, D., Belonogov, E., & Margarit, A. (2019, September 17). Optimization of Oil Field Development using a Surrogate Model: Case of Miscible Gas Injection. Society of Petroleum Engineers. doi:10.2118/196639-MS;

[27] Babin, V. M., Vashevnik, A. M., Ushmaev, O. S., Gruzdev, A. P., Pozdneev, A. V., Semenikhin, A. S., … Ciaurri, D. E. (2016, October 24). A Methodology for the Refinement of Well Locations During Operational Drilling in Presence of Geological Uncertainty. Society of Petroleum Engineers. doi:10.2118/181992-MS